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Scripting pwxcmd Commands for the PowerExchange Logger

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Abstract

This article describes how to issue *pwxcmd* commands from a script on a Windows or a UNIX machine to a PowerExchange Logger for Linux, UNIX, and Window process on a UNIX machine. This article applies to PowerExchange on Linux, UNIX, and Windows platforms. It includes example scripts.

Supported Versions

- PowerExchange 10.0 or later

Table of Contents

Overview. 2
Scenarios. 3
Task Flow. 3
Step 1. Configure the PowerExchange Logger to Receive pwxcmd Commands. 3
Step 2. Configure a Connection to a PowerExchange Logger to Send pwxcmd Commands. 4
Step 3. Restart the PowerExchange Logger on UNIX. 4
Step 4. Script pwxcmd Commands. 5
Example Script for pwxcmd for Windows. 5
Example Script for pwxcmd for UNIX. 6
Step 5. Run the pwxcmd Script. 7

Overview

This article summarizes the steps that are necessary to issue pwxcmd commands in a script on a Windows or a UNIX machine to a PowerExchange Logger for Linux, UNIX, and Windows process running on a UNIX machine. It includes example scripts.

You can use the pwxcmd program to send commands to a PowerExchange Listener or a PowerExchange Logger process. With the pwxcmd program, you can issue the same commands that you can issue from the command line when these PowerExchange processes are running in foreground mode. You can issue pwxcmd commands in batch jobs or files, in scripts, and also from the command line.

To issue pwxcmd commands, you must perform configuration tasks on the following machines:

- The machine where the PowerExchange process that you want to contact runs. This article describes an implementation that issues pwxcmd commands to a PowerExchange Logger for Linux, UNIX, and Windows.
- The machine from which you issue pwxcmd commands. In the implementation described in this article, the pwxcmd commands are issued from a script on a Windows or UNIX machine.

Note: The pwxcmd commands only work with the PowerExchange Logger for Linux, UNIX, and Windows and not with PowerExchange Condense on other platforms.

These instructions apply only to PowerExchange on Linux, UNIX, and Windows. Although this article focuses on Windows and UNIX, this information also applies to Linux

Scenarios

Issue pwxcmd commands in a script if they are required on a regular basis or if you need to issue the commands on a remote machine..

You can use a script to issue pwxcmd commands in the following circumstances:

- You need to issue some pwxcmd commands with specific options and arguments on a regular basis, such as to request a PowerExchange Logger for Linux, UNIX, and Windows process on a UNIX machine to perform a file switch.
- You want to issue the commands from a remote Windows or UNIX machine. You create either a Windows batch file or a UNIX bash shell script to issue one or more pwxcmd commands with options and arguments.

Task Flow

You must perform the following steps to send pwxcmd commands from a script on a Windows or UNIX machine to a PowerExchange Logger process on a UNIX machine:

1. "Step 1. Configure the PowerExchange Logger to Receive pwxcmd Commands" on page 3

   On the machine where the PowerExchange Logger runs, include the following configuration statements:

   - CONDENSENAME statement in the pwxccl.cfg file to configure a name for the command-handling service for the PowerExchange Logger process that will receive *pwxcmd* commands.
   - SVCNODE statement in the dbmover.cfg file to configure a TCP/IP port on which the command-handling service for the PowerExchange Logger process listens for *pwxcmd* commands.
2. "Step 2. Configure a Connection to a PowerExchange Logger to Send pwxcmd Commands" on page 4

   On the machine from which you will send pwxcmd commands, include a CMDNODE statement in the dbmover.cfg file.
3. "Step 3. Restart the PowerExchange Logger on UNIX" on page 4
4. "Step 4. Script pwxcmd Commands" on page 5
5. "Step 5. Run the pwxcmd Script" on page 7

Step 1. Configure the PowerExchange Logger to Receive pwxcmd Commands

To configure the PowerExchange Logger to receive pwxcmd commands, you must specify a CONDENSENAME statement in the pwxccl.cfg configuration file on the PowerExchange Logger for Linux, UNIX, and Windows machine. Additionally, you must specify a SVCNODE statement in the dbmover.cfg file on the PowerExchange Logger machine.

1. In the pwxccl.cfg configuration file on the PowerExchange Logger machine, add a CONDENSENAME statement to configure a name for the command-handling service for the PowerExchange Logger process that will receive pwxcmd commands.

   For example:

   ```
   /*****************************************************************/
   /*  pwxccl.cfg Configuration File
   /*****************************************************************/
   CONDENSENAME=COND_NODE
   ```

   For more information about the CONDENSENAME statement, see the *PowerExchange CDC Guide for Linux, UNIX, and Windows*.

2. In the dbmover.cfg file on the PowerExchange Logger machine, add a SVCNODE statement to configure a TCP/IP port on which the command-handling service for the PowerExchange Logger process listens for pwxcmd commands.

   For example:

   ```
   /*****************************************************************/
   /*  PowerExchange Configuration File /
   *****************************************************************/
   SVCNODE=(COND_NODE,6003)
   ```

   In the SVCNODE statement:

   - The first parameter is the service name. This service name must match the service name in the corresponding CONDENSENAME statement in the pwxccl.cfg file on the PowerExchange Logger machine.
   - The second parameter is the TCP/IP port number on which the command-handling service listens for pwxcmd commands. Specify this port number in the corresponding CMDNODE statement in the dbmover.cfg file on the machine from which pwxcmd commands are issued

Step 2. Configure a Connection to a PowerExchange Logger to Send pwxcmd Commands

In the dbmover.cfg file on the machine from which pwxcmd commands will be issued, specify a CMDNODE statement to configure connection information for a PowerExchange Logger for Linux, UNIX, and Windows to which you want to send pwxcmd commands.

For example:

```
/*****************************************************************/
/*  PowerExchange Configuration File
/*****************************************************************/
CMDNODE=(cndnode,CONDENSE,unix_host,6003)
```

In the CMDNODE statement:

- The first parameter is a service name for the pwxcmd program. This service name can be any name you want. Specify this service name when you issue pwxcmd commands to the PowerExchange Logger process.
- The second parameter is the service type, which must be CONDENSE for a PowerExchange Logger process.
- The third parameter is the name of the host name or IP address where the PowerExchange Logger process is running.
- The fourth parameter is the TCP/IP port number on which the service listens that matches the port number in the corresponding SVCNODE statement in the dbmover.cfg file on the PowerExchange Logger machine.

Step 3. Restart the PowerExchange Logger on UNIX

To issue pwxcmd commands to the PowerExchange Logger for Linux, UNIX, and Windows process, you must restart the PowerExchange Logger.

For example, issue the following command:

```
pwxccl
```

Note: You cannot use the pwxcmd program to start the PowerExchange Logger.

For more information, see the *PowerExchange Command Reference*.

Step 4. Script pwxcmd Commands

1. Copy one of the scripts provided in this article to the machine from which you want to issue pwxcmd commands.
   - Windows batch file: **pwxcmd_cnd.bat**. For more information, see "Example Script for pwxcmd for Windows" on page 5.
   - UNIX Perl script: **pwxcmd_cnd.pl** . For more information, see "Example Script for pwxcmd for UNIX" on page 6.
2. On each pwxcmd command in your script, use the following command syntax:

```
pwxcmd command_name
    -{service|sv} service_name
    [-{user|uid|u} user_ID]
    [-{epassword|e encrypted_password]
    [-{password|pwd|p} password]
    [command_options]
```

3. Modify the script to issue any of the following pwxcmd commands to a PowerExchange Logger process:

   displaystatus

   Displays the current status of the PowerExchange Logger Controller and each subtask.

   condense

   Starts a new PowerExchange Logger logging cycle.

   fileswitch

   Closes the current PowerExchange Logger log files if they contain data and then switches to a new set of log files. If the current log files do not contain any data, the file switch does not occur.

   shutcond

   Stops the PowerExchange Logger after performing a final logging cycle.

   shutdown

   Stops the PowerExchange Logger.

   The pwxcmd program sends the commands to the service specified in the **-service** parameter on the command. This service must match the service name value that is coded in the CMDNODE statement in the dbmover.cfg file that points to the PowerExchange Logger process that is the target of the command.

   Note: If you enable security for pwxcmd commands, you must specify a user ID and password or an encrypted password on the command. In this example, pwxcmd commands are issued without security enabled.

Example Script for pwxcmd for Windows

You can modify the following example UNIX bash shell script, pwxcmd_cnd.pl, to issue any of the included *pwxcmd* commands to a PowerExchange Logger process:

```
#!/usr/bin/bash
# Sample pwxcmd script - pwxcmd_cnd.pl
#
# Issue the following pwxcmd commands and exit on any error:
#
# displaystatus
# condense
# fileswitch
# shutcond
# shutdown
#
# Check status of PowerExchange Logger process
#
```

```
pwxcmd displaystatus -sv cndnode
if [ "$?" != 0 ]; then
# handle error
echo "Could not display status of PowerExchange Logger process"
exit
fi
# Start PowerExchange Logger logging cycle
pwxcmd condense -sv cndnode
if [ "$?" != 0 ]; then
# handle error
echo "Could not start PowerExchange Logger logging cycle process"
exit
fi
# Perform a file switch
pwxcmd fileswitch -sv cndnode
if [ "$?" != 0 ]; then
 handle error
echo "Could not perform file switch"
exit
fi
# Stop PowerExchange Logger after final logging cycle
pwxcmd shutcond -sv cndnode
if [ "$?" != 0 ]; then
# handle error
echo "Could not stop the PowerExchange Logger process"
exit
fi
# Stop PowerExchange Logger
pwxcmd shutdown -sv cndnode
if [ "$?" != 0 ]; then
# handle error
echo "Could not stop the PowerExchange Logger process"
exit
fi
```

Example Script for pwxcmd for UNIX

You can modify the following example Windows batch file, `pwxcmd_cnd.bat`, to issue any of the included pwxcmd commands to a PowerExchange Logger process:

```
@echo off
REM -- Sample pwxcmd script
REM --
REM -- Issue the following pwxcmd condense commands and exit on any error:
REM --
REM -- displaystatus
REM -- condense
REM -- fileswitch
REM -- shutcond
REM -- shutdown
REM --
REM -- Check status of PowerExchange Logger process
REM --
pwxcmd displaystatus -sv cndnode
REM -- handle error
if errorlevel ==32601 goto noconnect
if errorlevel ==0 goto condense
REM -- Start a PowerExchange Logger logging cycle
:condense
pwxcmd condense -sv cndnode
if %errorlevel% neq 0 goto noconnect
if errorlevel ==0 goto fileswitch
REM -- Perform a file switch
:fileswitch
pwxcmd fileswitch -sv cndnode
if %errorlevel% neq 0 goto noconnect
if errorlevel ==0 goto shutcond
REM -- Stop PowerExchange Logger after final logging cycle
:shutcond
```

```
pwxcmd shutcond -sv cndnode
if %errorlevel% neq 0 goto noconnect
if errorlevel ==0 goto end
REM -- Stop the PowerExchange Logger
:shutdown
pwxcmd shutdown -sv cndnode
if %errorlevel% neq 0 goto noconnect
if errorlevel ==0 goto end
:noconnect
echo %errorlevel%
echo "Could not connect to pwxcmd command handler."
pause
goto end
:end
```

Step 5. Run the pwxcmd Script

You can run the pwxcmd_cnd.bat file on Windows or the pwxcmd_cnd.pl script on a UNIX machine.

1. To run the pwxcmd script on Windows, perform the following steps:
   a. On the Windows machine where you created the batch file, change directory to the directory where you stored the script.
   b. At a command prompt, issue the following command:

      ```
      pwxcmd_cnd.bat
      ```
2. To run the pwxcmd script on Unix, perform the following steps:
   a. On the UNIX machine where you created the batch file, change directory to the directory where you stored the script.
   b. At a command prompt, issue the following command:

      ```
      . pwxcmd_cnd.pl
      ```

Authors

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